

Lebanese University – Faculty of Engineering

Subject: Sensors

Professor: Dr. Mohammad Aoude



**Sensors project
(Greenhouse)**

Presented by:

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1. Summary of Contribution

I led the calibration of every analog sensor in the system. For each sensor (DHT11, LDR, soil moisture, water level, sound, YF-S201, HC-SR04) I recorded raw readings against three known reference points and derived the linear calibration equations and thresholds documented in the final report.

I ran the 120-second continuous live test used in the experimental results section, logging minimum, maximum, and average values for temperature, humidity, flow rate, soil moisture, water level, ultrasonic distance, and sound level, and verified that every threshold-based automatic response (pump, fan, grow-LEDs, buzzer, sound alarm) triggered correctly during the test.

I also compiled the measurement uncertainty and error analysis, comparing each sensor's datasheet accuracy against its measured error and repeatability, and identified the main noise sources (self-heating, turbulence, EMI, sensor oxidation) for each.

2. Git Activity

- Add calibration test script for analog sensors
- Record DHT11 reference points and derive T_cal/H_cal
- Derive soil-moisture dry/wet threshold from 3-point test
- Derive YF-S201 pulse-to-flow conversion factor
- Log 120s continuous test data (temp, humidity, flow, soil)
- Add measurement uncertainty and error analysis notes
- Verify threshold-triggered actuator responses in test log

3. Reflection

Calibration turned out to be far more iterative than I expected — the first pass at the soil moisture threshold was off because I hadn't accounted for probe drift between tests. Running the full 120-second live test end-to-end was satisfying because it showed all four of our subsystems working together instead of in isolation. I came away with a much better appreciation for why datasheet accuracy numbers don't tell the whole story until you've measured repeatability yourself